git init , git remote , git reset , git stash , git fork

git init :

```
Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git
$ ls

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git
$ git init
Initialized empty Git repository in C:/Users/Mourian P/Downloads/git/.git/

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (master)
$ git remote -v

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (master)
$ git remote add origin https://github.com/Mourian-sys/git.git
```

Syntax

git init

Example (Local Repository)

Step 1: Create a project folder

mkdir myproject

cd myproject

Step 2: Initialize Git

git init

Output:

Initialized empty Git repository

Now myproject is a Git repository.

Using git init with GitHub (Complete Flow)

Step 1: Initialize Git locally

git init

Step 2: Create a file

touch index.html

Step 3: Add file to staging area

git add index.html

Step 4: Commit the file

git commit -m "Initial commit"

Connecting Local Repo to GitHub

Step 5: Create a repository on GitHub

- Go to GitHub
- Click **New Repository**
- Copy the repository URL

Example:

https://github.com/username/myproject.git

Step 6: Add GitHub as remote

git remote add origin https://github.com/username/myproject.git

origin = default name for remote repository

Step 7: Push code to GitHub

git branch -M main

git push -u origin main

Now our local project is uploaded to GitHub

Important Points (Exam / Interview)

- git init creates a **new empty Git repository**
- It is used **only once** per project
- It works **locally first**
- To upload to GitHub, we must use:
    - git remote
    - git push

git init is used to initialize a new Git repository in a project directory so that Git can track file changes and manage version history

git reset:

```
Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git reset --soft ^C

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git log --oneline e3da04e
e3da04e (HEAD -> main, origin/main) 4th
9375f94 3rd
2d5e00d 2nd
56607be 1st commit

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git reset --soft e3da04e

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git log --oneline
e3da04e (HEAD -> main, origin/main) 4th
9375f94 3rd
2d5e00d 2nd
56607be 1st commit

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git reset --soft 9375f94

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git log --oneline
```

```
MINGW64:/c/Users/Mourian P/downloads/git
9375f94 (HEAD -> main) 3rd
2d5e00d 2nd
56607be 1st commit

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git reset --mixed 2d5e00d

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git log --oneline
2d5e00d (HEAD -> main) 2nd
56607be 1st commit

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git reset --hard 2d5e00d
HEAD is now at 2d5e00d 2nd

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git reset --hard 56607be
HEAD is now at 56607be 1st commit

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git log --oneline
56607be (HEAD -> main) 1st commit

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ ls
file1  file2  file3  file5  file6

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git log --oneline
56607be (HEAD -> main) 1st commit

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ touch file7

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git stash
No local changes to save

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file5
        file6
        file7

nothing added to commit but untracked files present (use "git add" to track)

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (main)
$ git push origin main
To https://github.com/Mourian-sys/git.git
 ! [rejected]        main -> main (non-fast-forward)
error: failed to push some refs to 'https://github.com/Mourian-sys/git.git'
```

Git reset –hard <hash>

Git reset –mixed<hash>

Git reset –reset<hash>

Git stash :

---

git stash

Definition:
git stash is used to **temporarily save uncommitted changes** (modified files) without committing them, so you can work on something else.

What happens when we use git stash?

- Modified files are saved in a **temporary storage (stash)**
- Working directory becomes **clean**
- Changes are not lost and can be restored later

Command

git stash

Example

Step 1: Modify a file

nano index.html

Step 2: Save changes temporarily git stash

Get back the stashed changes

Apply and remove stash

git stash pop

Apply without removing git stash apply

Other Useful git stash Commands

git stash list # shows all stashes

git stash drop # deletes latest stash

git stash clear # deletes all stashes

Git remote :

```
Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git
$ ls

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git
$ git init
Initialized empty Git repository in C:/Users/Mourian P/Downloads/git/.git/

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (master)
$ git remote -v

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/downloads/git (master)
$ git remote add origin https://github.com/Mourian-sys/git.git
```

git remote is used to **connect a local Git repository with a remote repository** like GitHub.

Common Commands

git remote # show remote names

git remote -v # show remote URLs

git remote add origin <repo-URL> # add remote

git remote remove origin # remove remote